Cape Town Airbnb listing analysis

Motlatsi Seeko
Student Number: SKXMOT001

Course: Supervised learning

Department of Statistical Science

University of Cape Town

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1.section 1

1.1 Introduction

Short-term rental platforms such as Airbnb have raised concerns regarding housing affordability and rental availability in Cape Town. This project aimed to model the probability that an Airbnb listing would not be fully booked over the next 30 days, where the target class ("no") represents listings expected to remain vacant for at least one night. The provided dataset was prepared for modelling, while a brief exploratory data analysis (EDA) was performed to understand the underlying data structure. Several supervised learning models were then trained and evaluated using 10-fold cross-validation, model performance metrics were computed and compared, and the best performing model was selected for final interpretation and prediction.

1.2 Exploratory data analysis

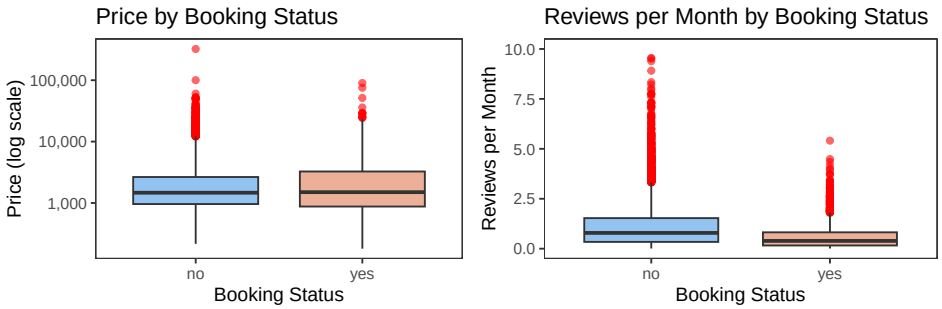


Figure 1: Boxplots showing Price (log scale) and Reviews per Month by Booking Status (fully_booked_30) for Airbnb listings in Cape Town

The two box plots as shown in Figure 1 offer a preliminary look at how price and review activity relate to booking status. In the Price by Booking Status plot, both the "yes" (fully booked) and "no" (not fully booked) groups are centred around a similar median of roughly R1,000–R2,000 per night, suggesting price alone is unlikely to be a strong discriminator; however, the "no" group displays a longer upper tail with more extreme outliers, hinting that very high-priced listings tend to remain vacant, a reasonable finding given that premium properties attract a narrower market. Turning to the Reviews per Month by Booking Status plot, the "no" group shows a noticeably higher median and greater spread compared to the fully booked "yes" group, which is somewhat counterintuitive but may reflect that listings with higher guest turnover (shorter stays, more frequent bookings) accumulate reviews faster while still experiencing occasional vacant nights, whereas fully booked listings may cater to longer-stay guests who review less frequently.

2. Section 2

2.1 Model Training

2.1.1. Vanilla Logistic regression

In the vanilla logistic regression, a number of variables emerge as notably significant.In the vanilla logistic regression, a number of variables emerge as notably significant. The strongest predictors with p-values effectively at zero are reviews_per_month, host_listings_count, and host_years, suggesting that a host's experience, portfolio size, and the frequency of guest reviews are the most influential drivers of whether a listing is not fully booked. host_response_time within a day is similarly highly significant, indicating that response behavior plays a meaningful role. At the 1% level, review_scores_rating and instant_bookable are also significant, implying that listing quality and booking convenience matter to occupancy. A further group host_acceptance_rate, host_response_time within a few hours, bedrooms, latitude, and longitude, are significant at the 5% level, suggesting that geographic location and physical listing characteristics carry modest but real predictive value. Conversely, several variables appear notably non-significant. beds ($p = 0.995$), maximum_nights ($p = 0.829$), accommodates ($p = 0.797$), number_of_reviews ($p = 0.664$), and minimum_nights ($p = 0.613$) all show p-values far exceeding any conventional threshold, indicating negligible marginal contribution once other variables are accounted for host_is_superhost, bathrooms, room_type, and the night-average variables similarly fail to reach significance.

2.2.1. Lasso regularization

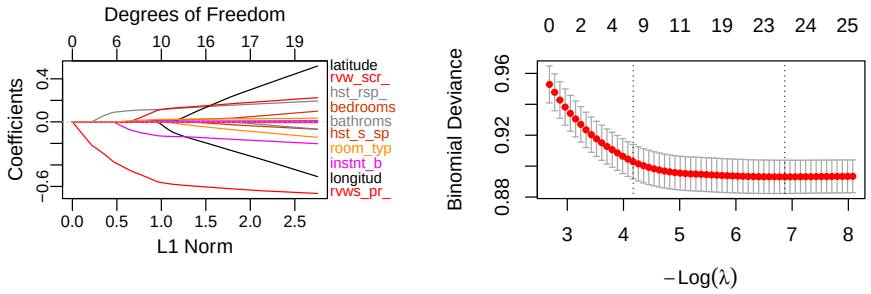


Figure 2: Lasso Regularization Path and Cross-Validation Deviance for L1-Regularised Logistic Regression

Figure 2 The Lasso coefficient paths (left) show that reviews_per_month and latitude enter the model earliest and attain the largest magnitudes, identifying them as the dominant predictors,

while most other variables enter later or remain near zero under strong regularization. The CV deviance curve (right) flattens beyond .min, confirming that meaningful predictive signal is concentrated in a small subset of variables. Based on the left dashed line, .min is selected, retaining 9 variables at minimum deviance while shrinking the remainder to exactly zero.

Table 1: Retained Variables under Lambda.min (L1-Regularised Logistic Regression)

Variable	Coefficient
reviews_per_month	-0.6540
longitude	-0.4410
latitude	0.4290
review_scores_rating	0.2087
instant_bookablef	0.1944
host_response_timewithin a day	0.1911
host_response_timewithin an hour	-0.1723
room_typeEntire home/apt	0.1269
bedrooms	0.0809
host_is_superhostf	0.0571
bathrooms	-0.0495
host_years	0.0330
host_response_timea few days or more	0.0102
minimum_nights_avg_ntm	0.0097
host_listings_count	-0.0067
host_acceptance_rate	0.0035
host_response_rate	0.0004
number_of_reviews	-0.0002
maximum_nights_avg_ntm	-0.0001
price	0.0000
maximum_nights	0.0000
instant_bookablet	0.0000
host_is_superhostt	0.0000
room_typePrivate room	0.0000

@tab-retained presents the variables retained under .min from the L1-regularized logistic regression. .min was selected over .1se as it achieves the lowest cross-validation deviance and retains a richer set of predictors, which is preferable given that log-loss, the evaluation metric rewards well-calibrated probability estimates.

2.3.1. Random forest

The random forest was tuned using a full grid search over three hyperparameters: the number of variables randomly sampled at each split (mtry) was searched over (2, 4, 6, 8), the splitting rule (splitrule) was evaluated over Gini impurity and Hellinger distance, and the minimum node size (min.node.size) was varied over (1, 5, 10), yielding 24 combinations in total. Each combination was evaluated using 10-fold cross-validation with log-loss as the selection metric, and the number of trees was fixed at 300 throughout. The combination minimizing average CV log-loss was selected as the final model configuration

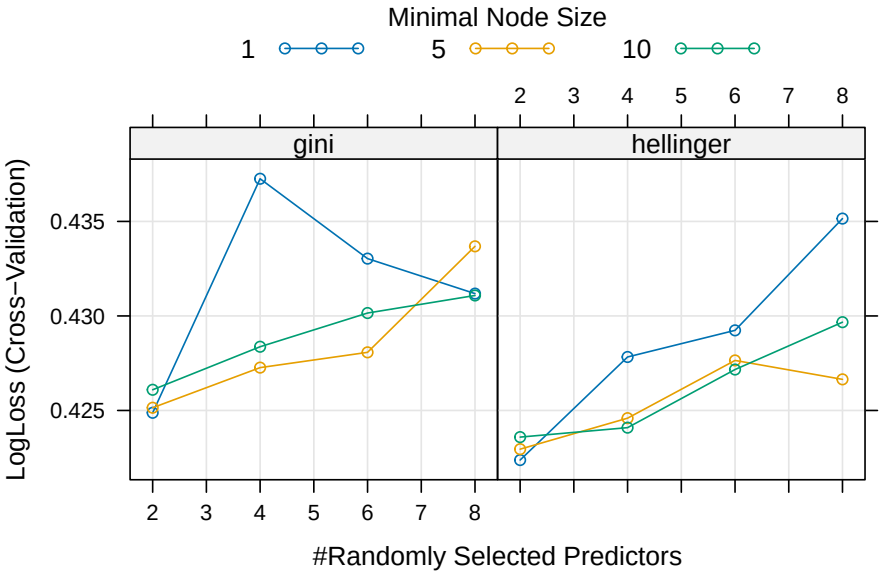


Figure 3: 10-Fold Cross-Validation Log-Loss for Random Forest Hyperparameter Grid Search

As shown in Figure 3 Hellinger splitting rule consistently achieves lower CV log-loss than Gini across all hyperparameter combinations. Within the Hellinger panel, lower mtry values paired with smaller node sizes yield the best performance. The final selected configuration is mtry = 2, splitrule = Hellinger, and min.node.size = 1, which achieved the minimum CV log-loss across all 24 combinations evaluated.

2.3.1. Boosting (GBM)

The GBM was tuned via a grid search over four hyperparameters, the number of boosting iterations (n.trees) was searched over {100, 200, 500}, interaction.depth (tree complexity) over

{1, 3, 5}, and the learning rate (shrinkage) over {0.05, 0.1}, yielding $3 \times 3 \times 2 = 18$ combinations. The minimum observations per node (n.minobsinnode) was fixed at 10 throughout. Each combination was evaluated using 10-fold CV with log-loss as the selection metric, and the combination minimizing average CV log-loss was selected as the final configuration.

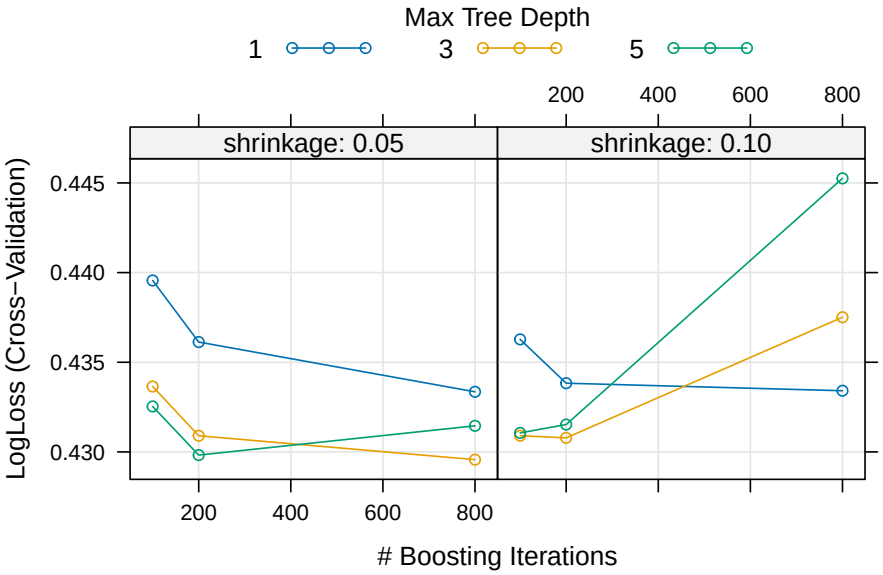


Figure 4: 10-Fold Cross-Validation Log-Loss for GBM Hyperparameter Grid Search

Figure 4 indicate that the lower shrinkage rate of 0.05 generally outperforms 0.10, producing more stable and lower log-loss values across boosting iterations. Within the shrinkage = 0.05 panel, intermediate tree depth (interaction.depth = 3) combined with more boosting iterations consistently achieves the lowest log-loss, while depth = 1 performs poorest throughout. The final selected configuration is n.trees = 500, interaction.depth = 3, shrinkage = 0.05, and n.minobsinnode = 10, confirming that a slower learning rate with deeper trees and more iterations yields the best CV performance.

Table 2: comparison of cross-Validation Performance Metrics Across All Models

	Model	LogLoss	AUPRC	Recall	Specificity	Precision	F1	Accuracy
Sensitivity...1	GLM	0.4434	0.9033	0.5672	0.7088	0.8966	0.6949	0.5932
Sensitivity...2	LASSO	0.4436	0.9032	0.5591	0.7105	0.8958	0.6885	0.5869
Sensitivity...3	Random Forest	0.4224	0.9149	0.6594	0.6805	0.9019	0.7618	0.6633
Sensitivity...4	GBM	0.4296	0.9107	0.7023	0.6276	0.8936	0.7865	0.6886

3. Section 3

a) Compere the models

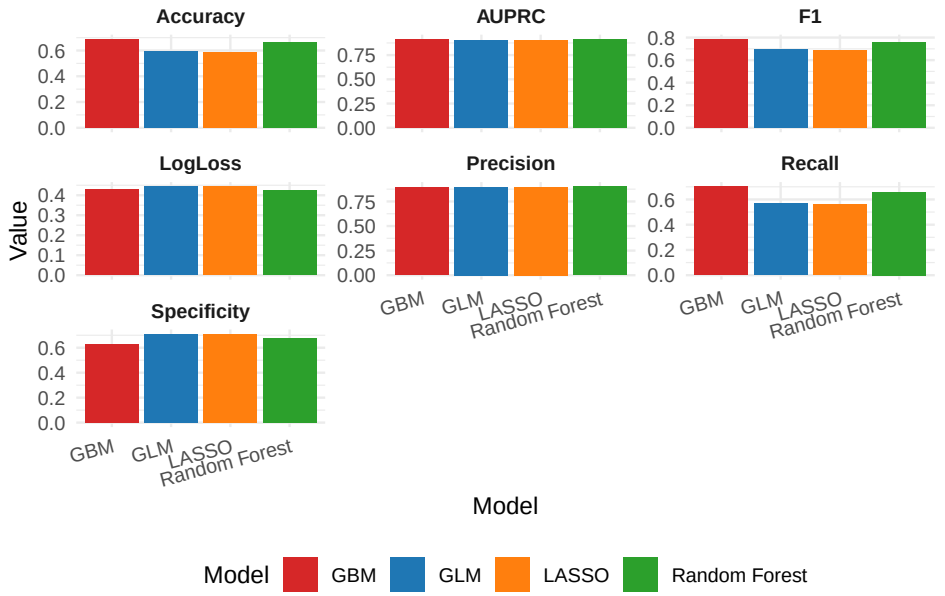


Figure 5: Visual Comparison of 10-Fold Cross-Validation Performance Metrics Across All Fitted Models

b) Best Model

The GBM model was selected as the final model because it achieved the best overall classification performance across the evaluation metrics as shown in @tab-com. It produced the highest Recall, F1-score, and Accuracy, indicating a strong ability to correctly identify listings that are not fully booked while maintaining a good balance between precision and sensitivity. Although the Random Forest model achieved slightly better Log-loss and AUPRC values,

GBM provided the most balanced and practically useful predictive performance under the chosen decision threshold, making it the most suitable model for this application and further illustration is indicated in Figure 5.

c) ROC Curve

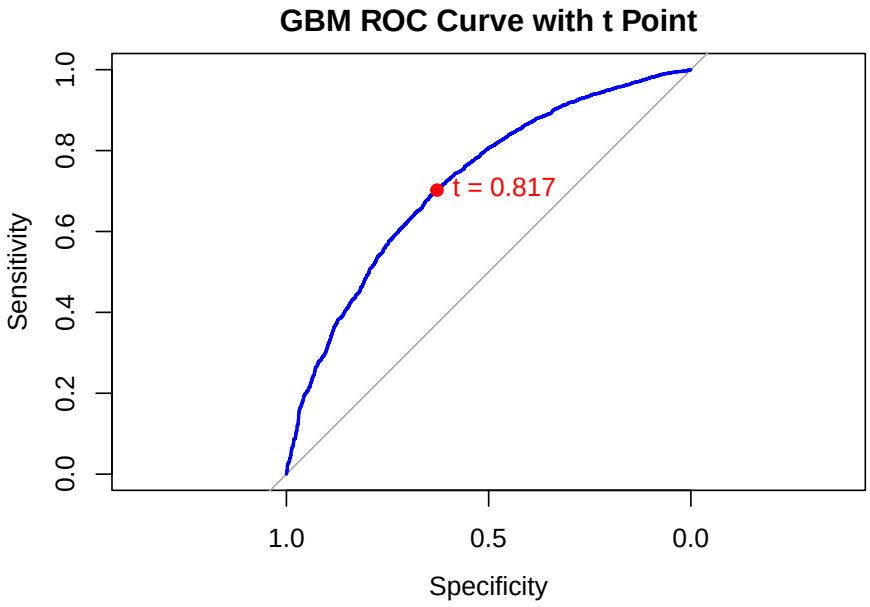


Figure 6: ROC Curve for the Selected GBM Model Indicating the Decision Threshold $= 0.817$

The ROC curve as indicated in Figure 6 for the GBM model demonstrates strong discriminating ability, curving well above the diagonal random-chance baseline. The red point marks the operating threshold at $= 0.817$, the proportion of target class observations corresponding to a sensitivity of approximately 0.70 and specificity of approximately 0.63, reflecting a reasonable balance between correctly identifying non-fully-booked listings and avoiding false positives.

4. Section 4

4.1 variable importance

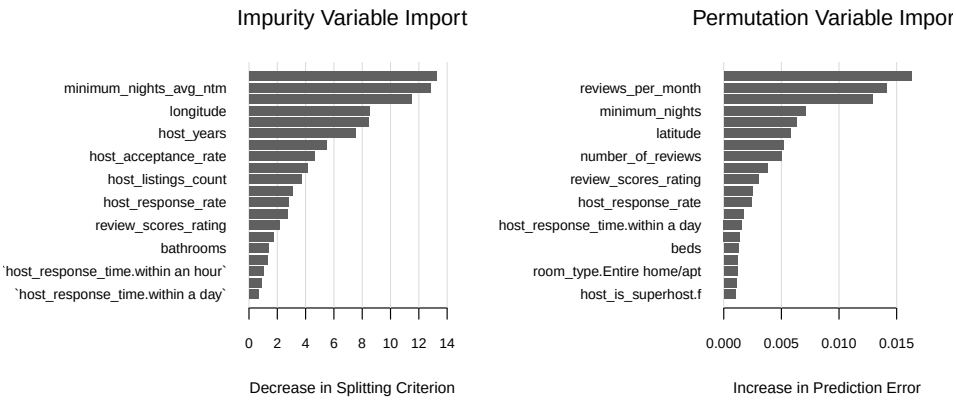


Figure 7: GBM Variable Importance: Impurity vs Permutation-Based Evaluation

As shown in Figure 7 Price, review activity, and minimum stay constraints are the strongest predictors of full-booking probability, appearing at the top of both plots. Geographic coordinates rank higher on impurity than permutation, suggesting their importance is slightly inflated by the splitting criterion. Both methods agree at the bottom, with bedrooms, bathrooms, and room type dummies consistently contributing little predictive value.

4.2 Partial dependency

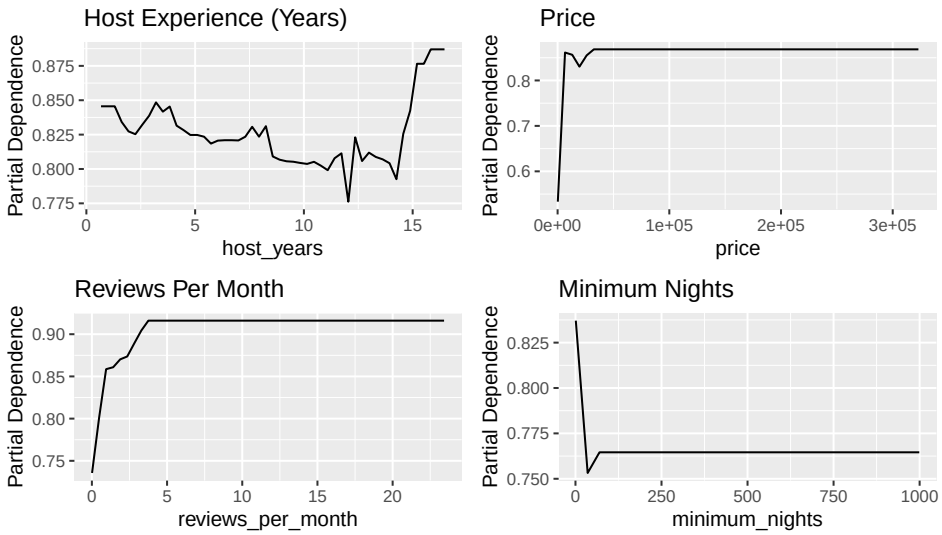


Figure 8: Partial Dependence Plots for Selected Predictors in the GBM Model.

The partial dependence plots in Figure 8 reveal nonlinear effects across key predictors. Higher prices and greater review activity are both associated with increased vacancy probability, while stricter minimum-night requirements initially reduce booking likelihood before stabilizing. Host experience shows a generally declining trend, though with some inconsistency across more experienced hosts.

5. Section 5

5.1 Predictions